

SAP CPI Project Documentation

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Scenario: Product Shipment Orders Processing

1.Scenario Overview

This integration handles **bulk shipment orders** by splitting them into individual orders, validating quantities, routing them based on region, and converting them into structured formats for warehouse systems. The scenario demonstrates the use of **Splitter**, **Filter**, **Router**, and **Converter** palette steps in SAP CPI and helps streamline logistics automation.

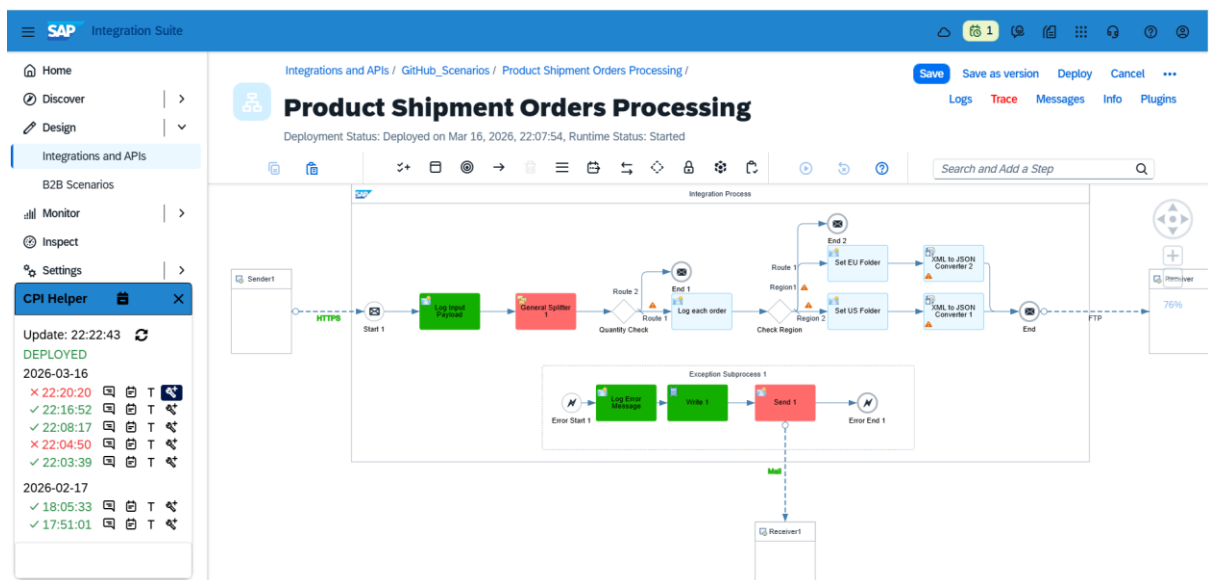
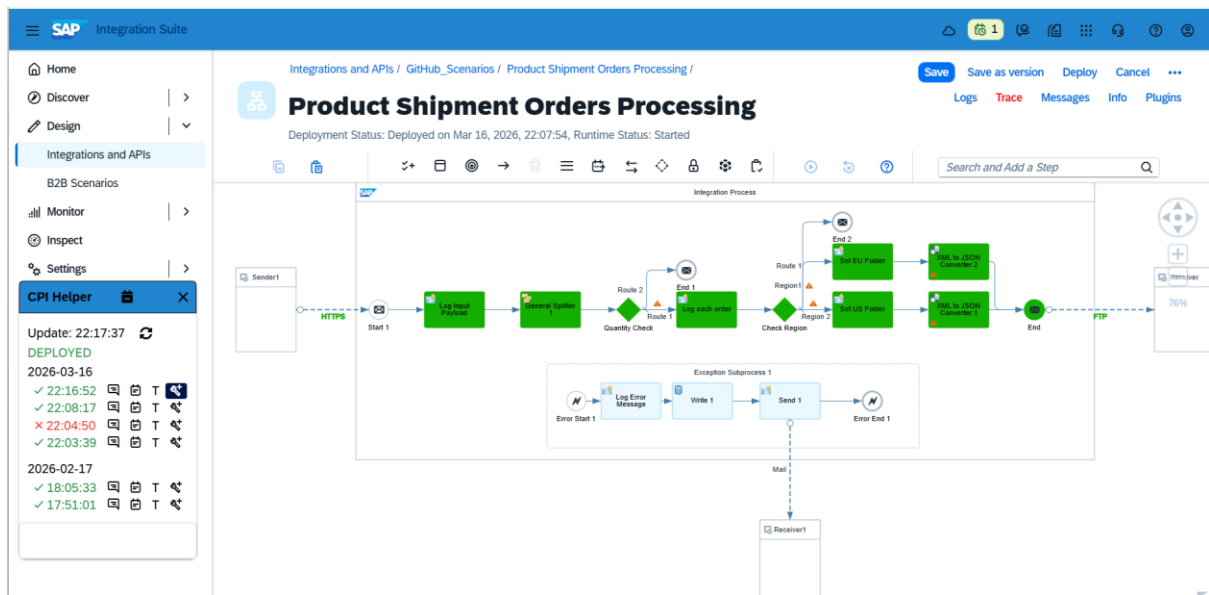
Business Use Case

A logistics system sends a **daily shipment file** containing multiple product orders for different regional warehouses (EU, US). Each order includes the product ID, quantity, and region.

Processing needs to:

- Validate orders (quantity > 0)
- Route based on region
- Convert format from XML to JSON
- Store orders as separate files in **region-specific folders**

2.Architechture Diagram



3. Integration Flow

Steps

1. HTTP picks up batch XML file

HTTPS	
General	Connection
REQUEST PROCESSING	
Address:	/ProductShipment/Orders
Authorization:	User Role
User Role:	ESBMessaging.send
CSRF Protected:	☒

2. **XML Splitter** splits each <Order> from the batch
3. **Filter** removes invalid orders (e.g., quantity = 0)
4. **Content Modifier** enriches each message with metadata (e.g., processed timestamp)
5. **Router** directs orders to regional branches (EU/US)

Router

General **Processing**

ERROR HANDLING

Throw Exception: ☐

ROUTING CONDITION

Order	Route Name	Condition Expression	Default Route
1	Route 2		☒
2	Route 1	//Quantity > '0'	☐

Router

General **Processing**

ERROR HANDLING

Throw Exception: ☐

ROUTING CONDITION

Order	Route Name	Condition Expression	Default Route
1	Region 2	//Region = 'US'	☐
2	Region1	//Region = 'EU'	☐
3	Route 1		☒

6. **XML to JSON Converter** transforms each valid order
7. **File Receiver Adapter** writes to /shipments/EU/ or /shipments/US/

8. Sample Input XML (Batch Shipment File)

<Shipments>

<Order>

<ID>ORD001</ID>

```
<Region>EU</Region>
<Quantity>10</Quantity>
</Order>
<Order>
  <ID>ORD002</ID>
  <Region>US</Region>
  <Quantity>0</Quantity>
</Order>
<Order>
  <ID>ORD003</ID>
  <Region>EU</Region>
  <Quantity>5</Quantity>
</Order>
</Shipments>
```

Processing Example

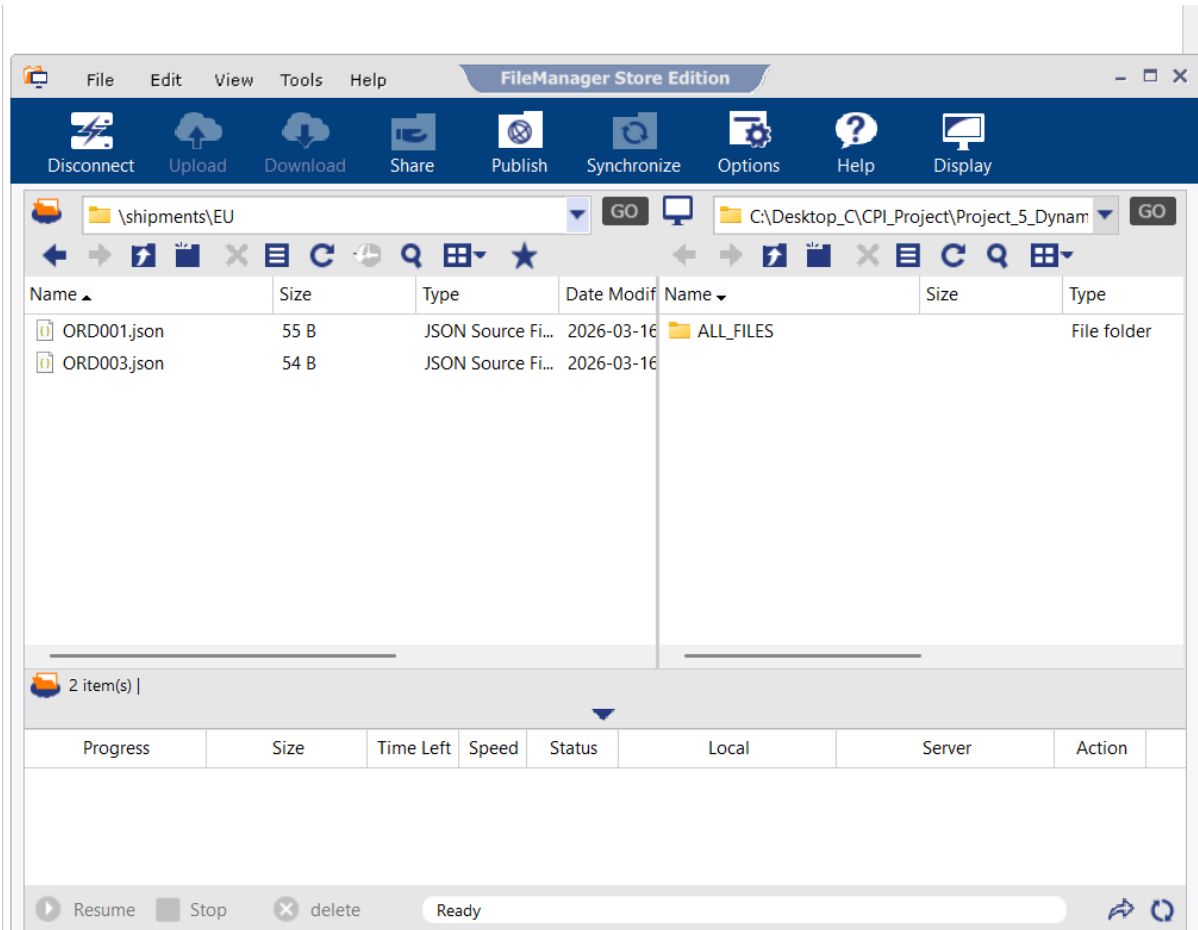
- **Filtered out:** ORD002 (Quantity = 0)
- **Processed and routed:** ORD001 → EU folder, ORD003 → EU folder

9. JSON Output Example (ORD001)

```
{
  "Order": {
    "ID": "ORD001",
    "Region": "EU",
    "Quantity": 10
```

}

}



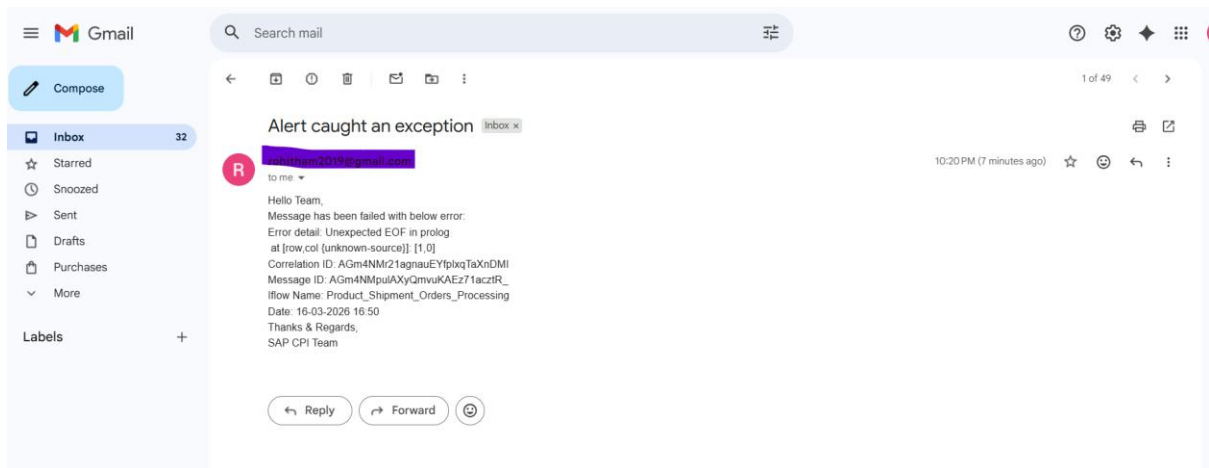
Files stored in HR system

File Output (File Receiver Adapter)

- Output path: /shipments/EU/ORD001.json /shipments/EU/ORD003.json
- Output format: JSON
- File naming: Based on order ID and region

10. Exception Handling

- Catch step errors in file write or transformation
- Log to Data Store
- Notify logistics support team via Gmail Adapter



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Iflow Package



Product Shipment Orders Processing.zip